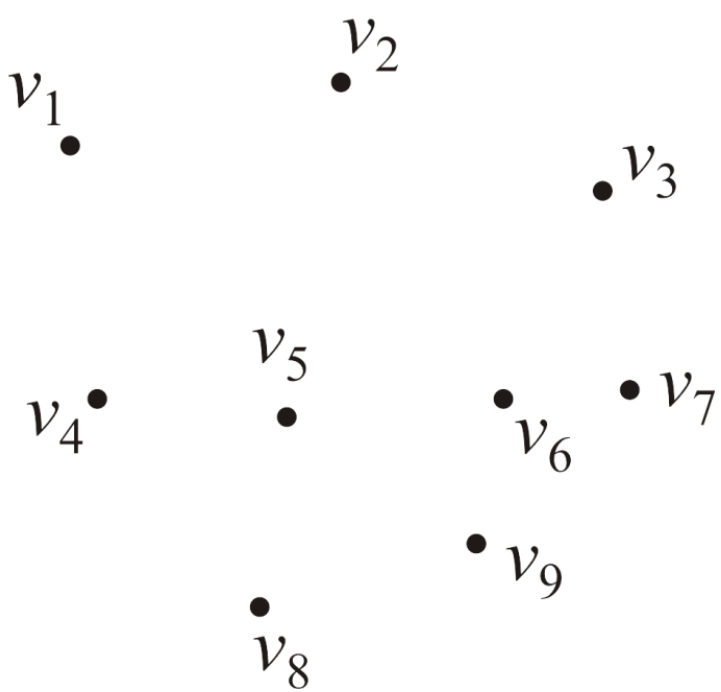
Data Structures

15. Graphs

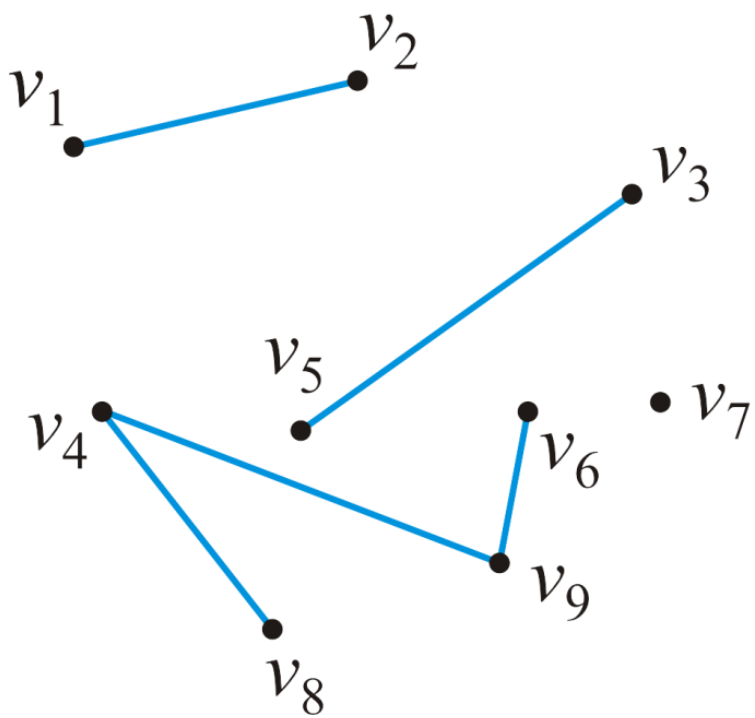
Graphs

- Consider this collection of vertices
  - $V = \{v_1, v_2, \ldots, v_9\}$
  - Where $|V| = n$



Graphs

- Associated with these vertices are $|E| = 5$ edges
  - $E = \{\{v_1, v_2\}, \{v_3, v_5\}, \{v_4, v_8\}, \{v_4, v_9\}, \{v_6, v_9\}\}$
- Pair $\{v_j, v_k\}$ indicates following relations
  - Vertex v_j is adjacent to vertex v_k
  - Vertex v_k is adjacent to vertex v_j



Applications Of Graphs

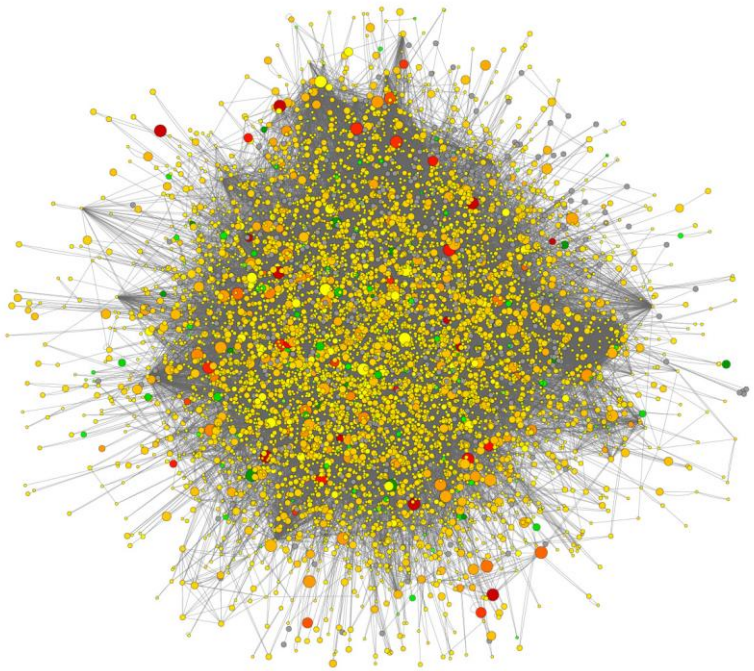
- Driving Map
  - Vertex = Intersection, destinations
  - Edge = Road
- Airline Traffic
  - Vertex = Cities serviced by the airline
  - Edge = Flight exists between two cities
- Computer networks
  - Vertex = Server nodes, end devices, routers
  - Edge = Data link

Applications Of Graphs

- Many real-world applications concern large graphs

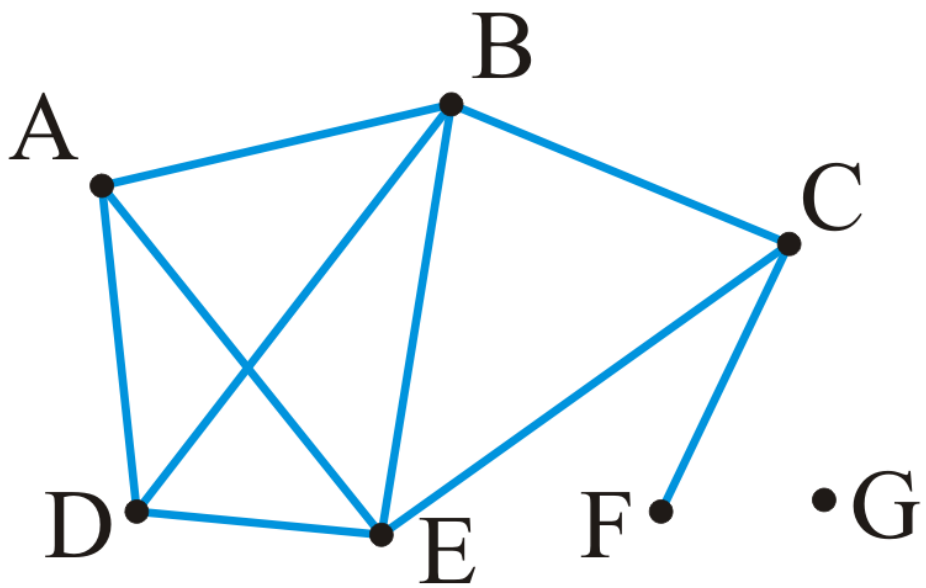
- Web document graph - 1 trillion webpages
  - Vertex = Webpage
  - Edge = Hyperlink

- Social networks - 1.3 billion users
  - Vertex = Users
  - Edge = Friendship relation



Undirected Graphs – Example

- Given |V| = 7 vertices and |E| = 9 edges
  - V = {A, B, C, D, E, F, G}
  - E = {{A, B}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, E}, {C, F}, {D, E}}



Undirected Graphs – Definition

- An undirected Graph is defined as $G=(V,E)$ consisting of
  - Set V of vertices: $V = \{v_1, v_2, \ldots, v_n\}$
    - Number of vertices is denoted by $|V| = n$
  - Set E of unordered pairs $\{v_i, v_j\}$ termed edges
    - Edges connect the vertices

- Maximum number of edges in an undirected graph is $O(|V|^2)$

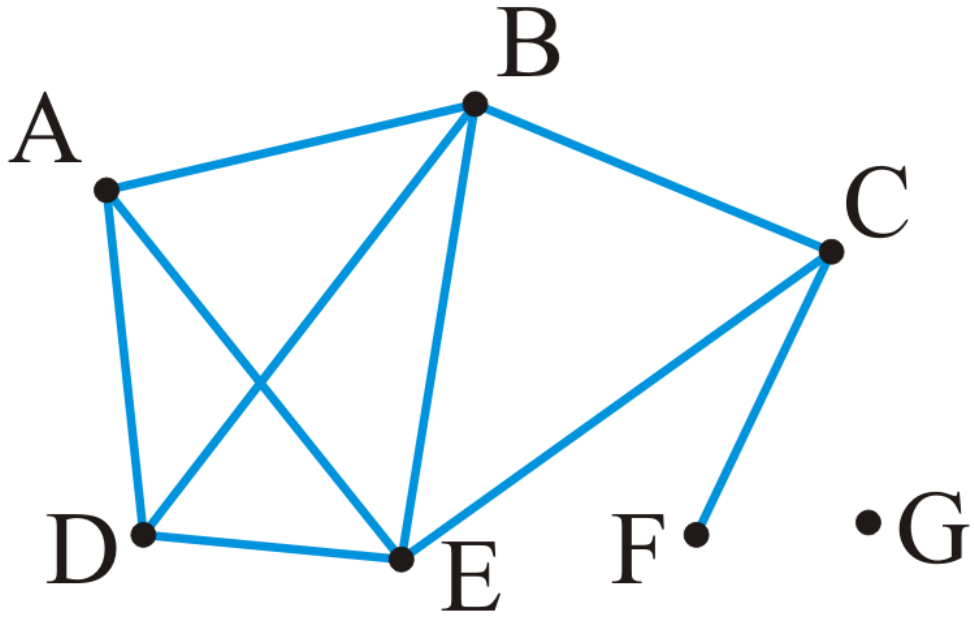
$$|E| \le \binom{|V|}{2} = \frac{|V|(|V|-1)}{2} = O\left(|V|^2\right)$$

  - Assumption: A vertex is never adjacent to itself
  - For example, $\{v_1, v_1\}$ will not define an edge

- Many data structures can implement abstract undirected graphs
  - Adjacency matrices, Adjacency lists

Degree

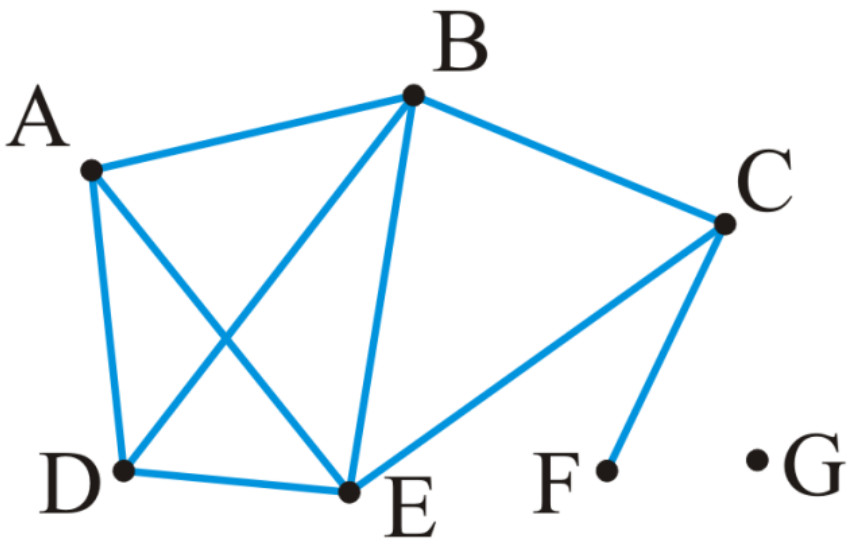
- Degree of a vertex is defined as the number of adjacent vertices
  - degree(A) = degree(D) = degree(C) = 3
  - degree(B) = degree(E) = 4
  - degree(F) = 1
  - degree(G) = 0

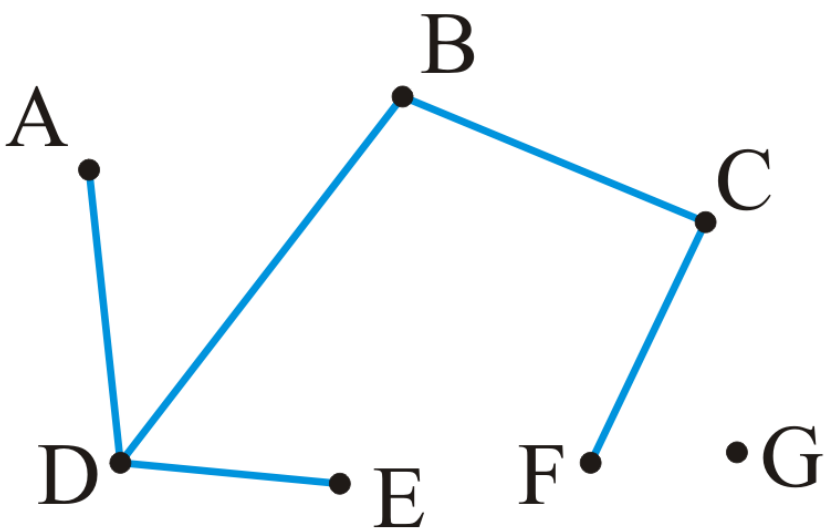


- Vertices adjacent to a given vertex are its neighbors

Subgraph

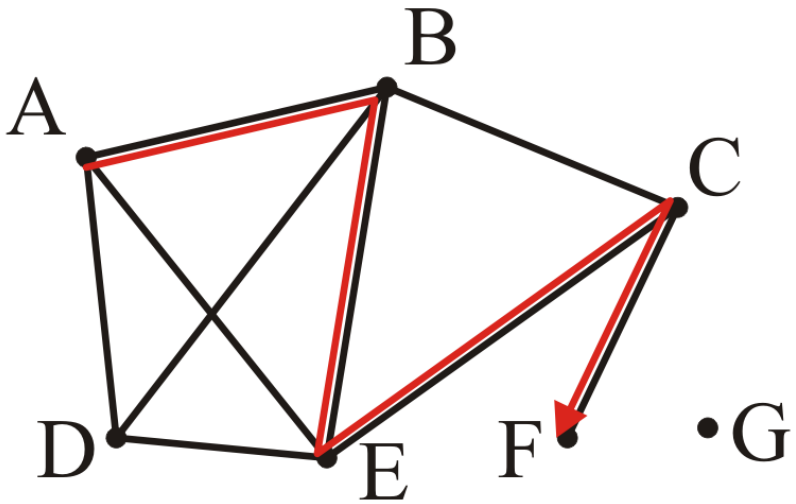
- A sub-graph of a graph G is defined by
  - Subset of the vertices
  - Subset of the edges that connected the subset of vertices in the original graph
- Every graph is a subgraph of itself





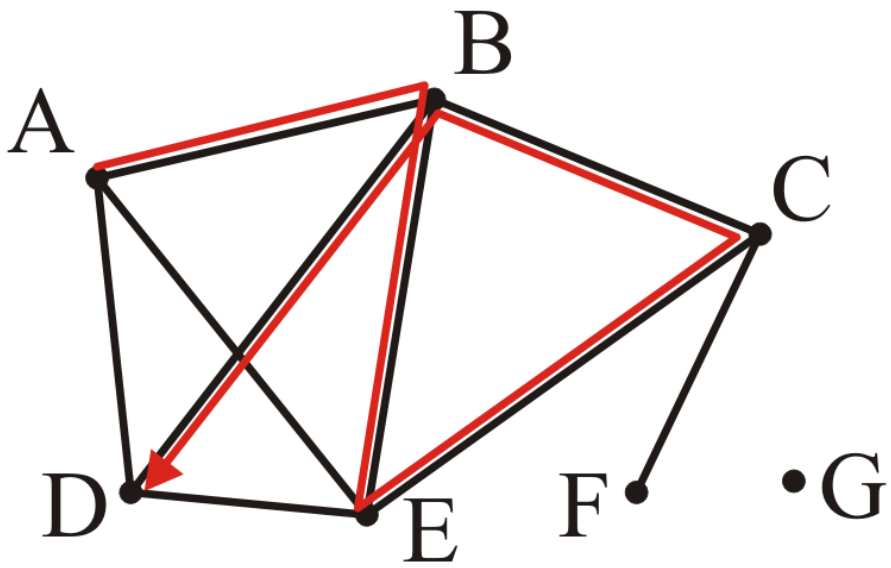
Path

- Path in an undirected graph is an ordered sequence of vertices
  - Consecutive vertices are connected through edges
- Path from vertex 0 to vertex k is $(v_0, v_1, v_2, \ldots, v_k)$
  - where $\{v_{j-1}, v_j\}$ is an edge for $j = 1, \ldots, k$
- Length of a path is equal to the number of edges
- Example: Path from A to F
  - Path: (A, B, E, C, F)
  - Length of the path is 4



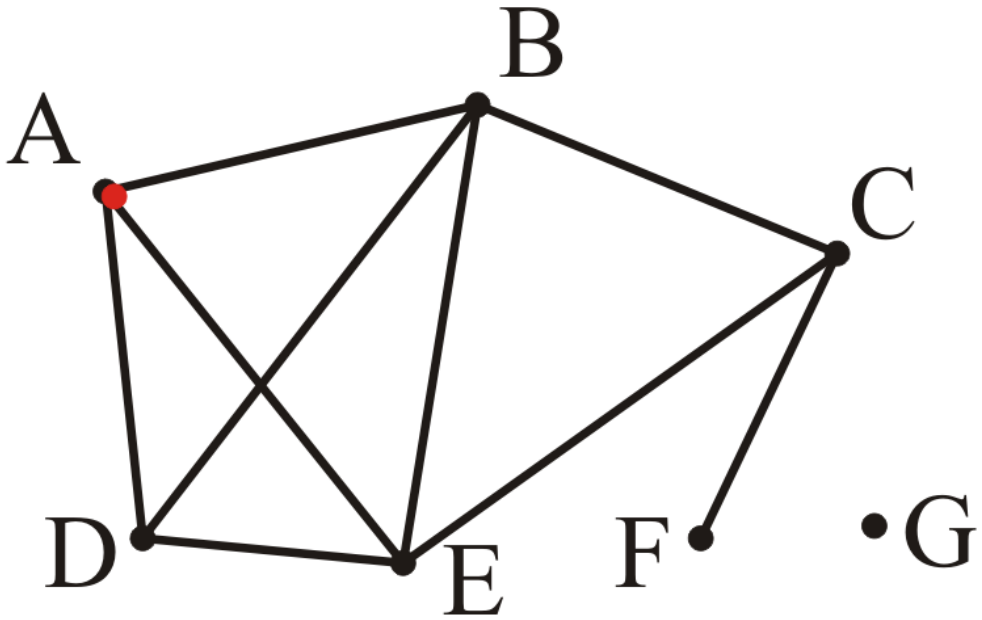
Path – Example

- Path of length 5: `(A, B, E, C, B, D)`
  - Repitition of vertex B



Path – Example

- A trivial path of length 0: `(A)`

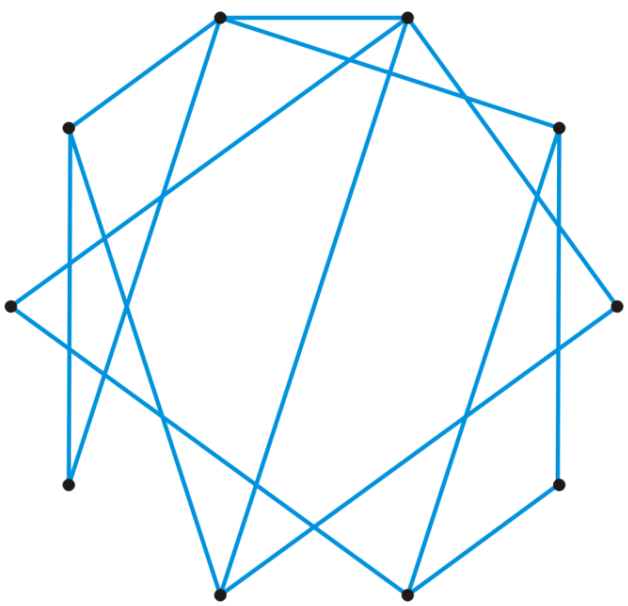


Simple Path

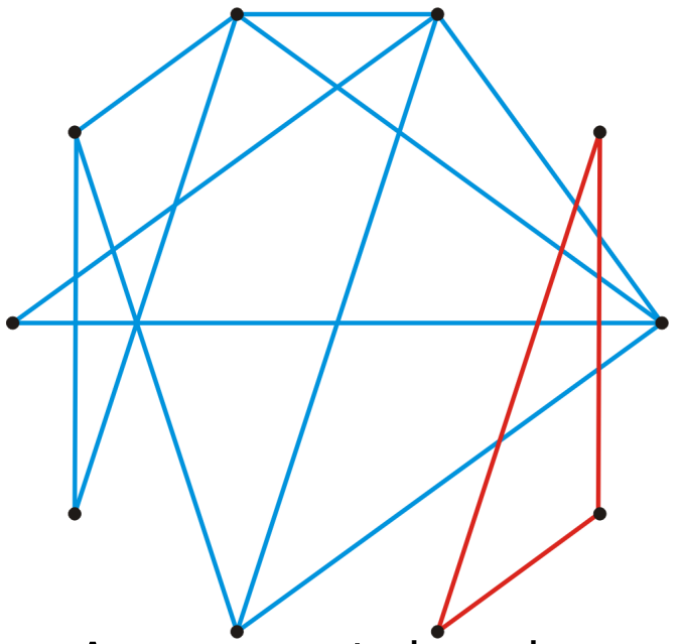
- A simple path has no repetitions other than perhaps the first and last vertices
- A simple cycle is a simple path of at least two vertices with the first and last vertices equal
  - Note: these definitions are not universal
- A loop is an edge from a vertex onto itself

Connectedness

- Two vertices v_i, v_j are said to be connected if there exists a path from v_i to v_j
- A graph is connected if there exists a path from every vertex to every other vertex



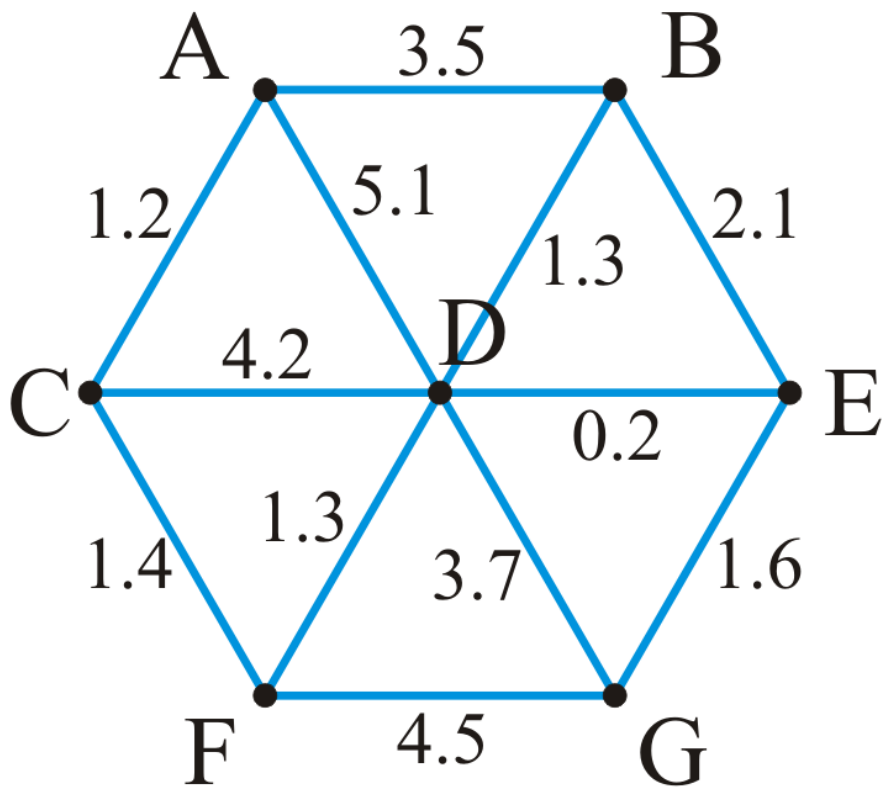
A connected graph



An unconnected graph

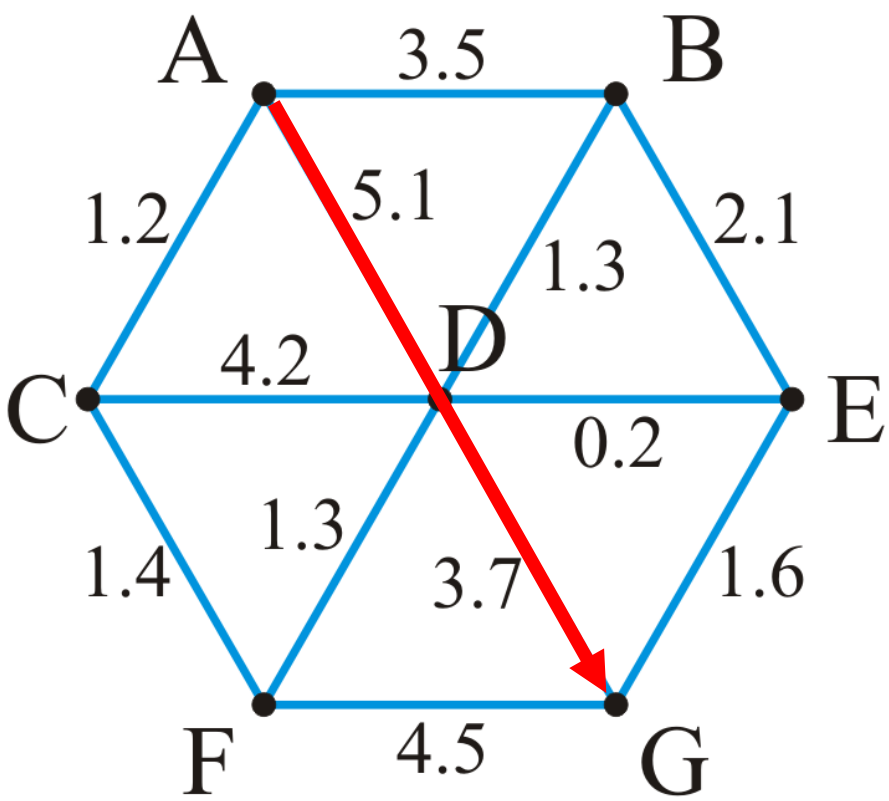
Weighted Graphs

- A weight may be associated with each edge in a graph
  - This could represent distance, energy consumption, cost, etc.
  - Such a graph is called a weighted graph
- Pictorially, we will represent weights by numbers next to the edges



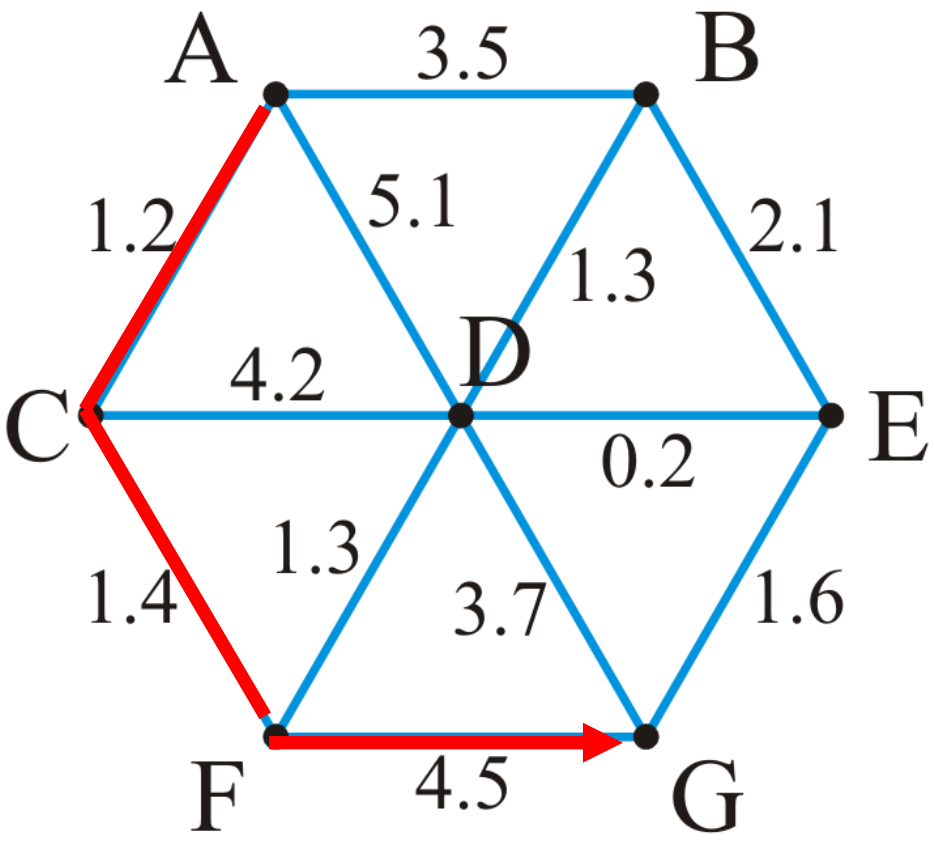
Weighted Graphs

- Length of a path within a weighted graph is the sum of all of the edges which make up the path
- The length of the path `(A, D, G)` in the following graph is `8.8`



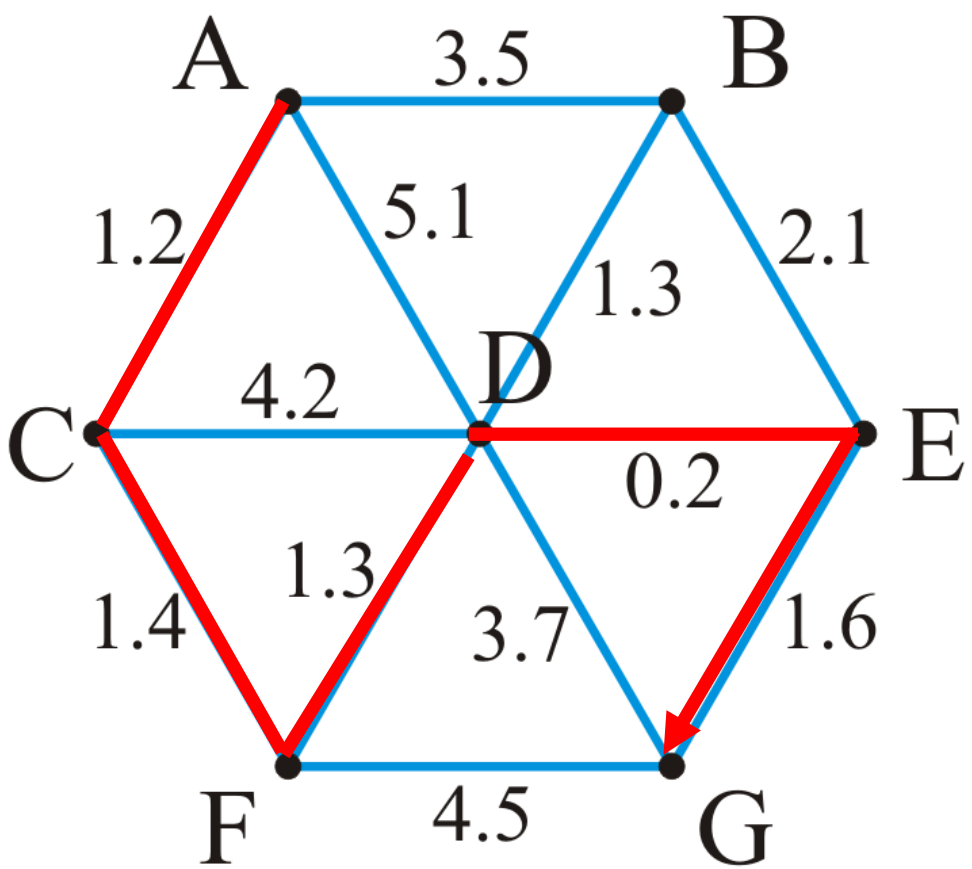
Weighted Graphs – Example

- Different paths may have different weights
  - Another path is `(A, C, F, G)` with length `1.2 + 1.4 + 4.5 = 7.1`



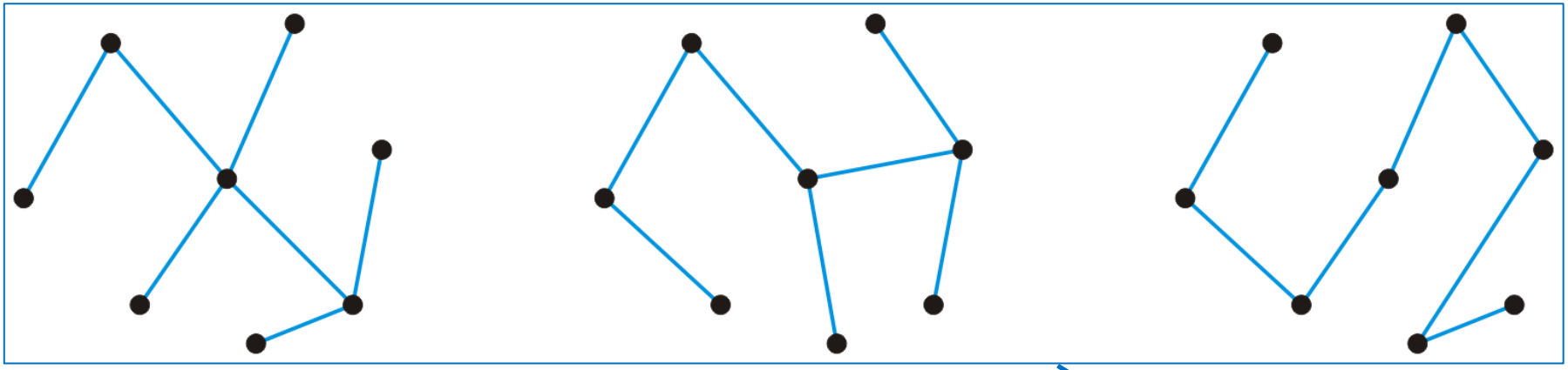
Weighted Graphs – Example

- Find the shortest path between two vertices A and G
- Shortest path is `(A, C, F, D, E, G)` with length `5.7`



Trees

- A graph is a tree if it satisfies the following two conditions
  - Graph is connected
  - There is a unique path between any two vertices



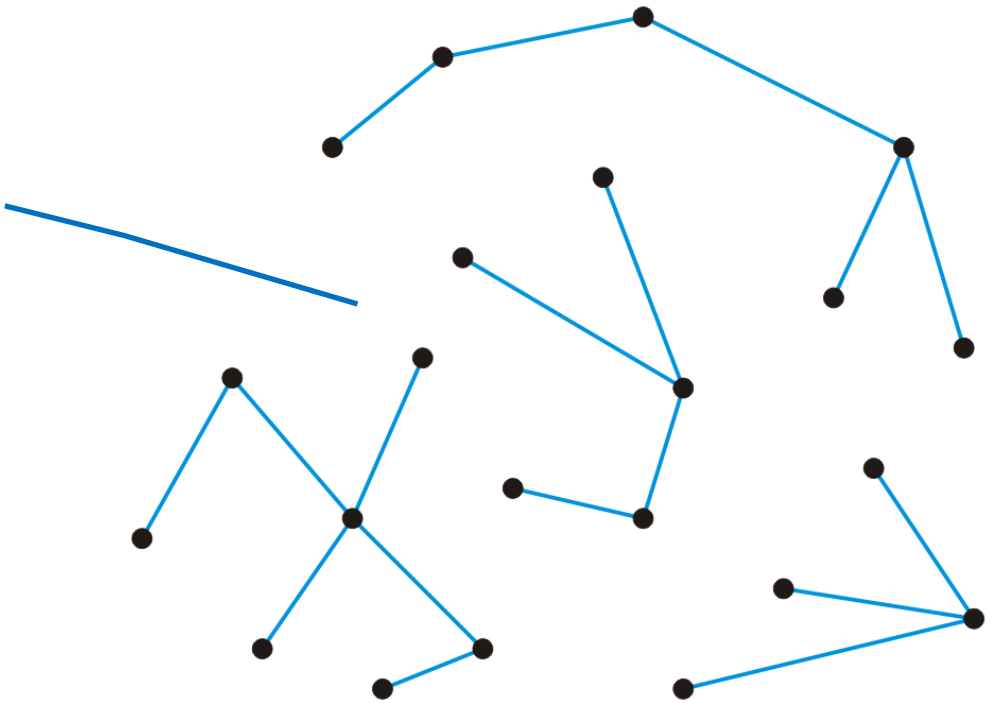
Three trees on same 8 vertices

- Consequences
  - The number of edges is $|E| = |V| - 1$
  - The graph is acyclic, that is, it does not contain any cycles
  - Adding one more edge must create a cycle
  - Removing any one edge creates two disjoint non-empty sub-graphs

Forest

- A forest is any graph that has no cycles
- Consequences
  - The number of edges is $|E| < |V|$
  - The number of trees is $|V| - |E|$
  - Removing any one edge adds one more tree to the forest

- Forest with 22 vertices and 18 edges
- Four trees

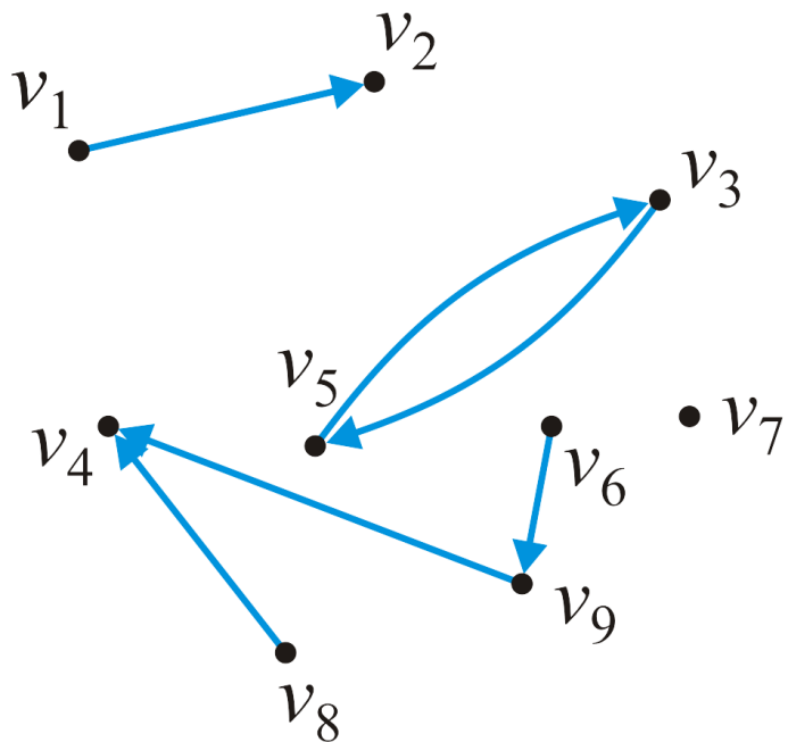


Directed Graphs

- In a directed graph, the edges on a graph are associated with a direction
  - Edges are ordered pairs (v_j, v_k) denoting a connection from v_j to v_k
  - The edge (v_j, v_k) is different from the edge (v_k, v_j)

Directed Graphs

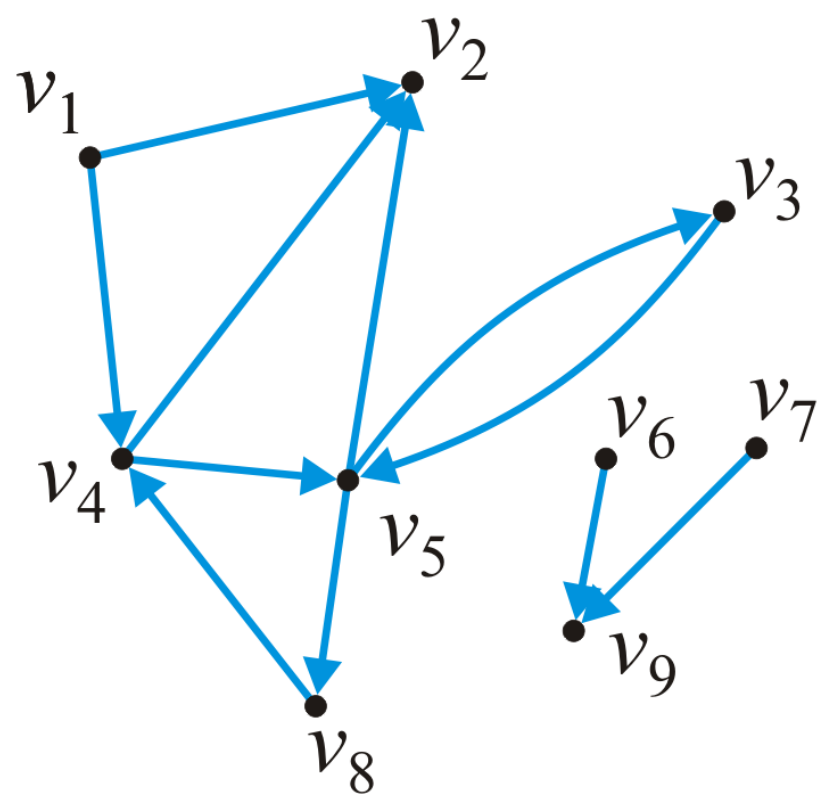
- Given our graph of nine vertices $V = \{v_1, v_2, \ldots v_9\}$
  - These six pairs (v_j, v_k) are directed edges
  - $E = \{(v_1, v_2), (v_3, v_5), (v_5, v_3), (v_6, v_9), (v_8, v_4), (v_9, v_4)\}$



In and Out Degree

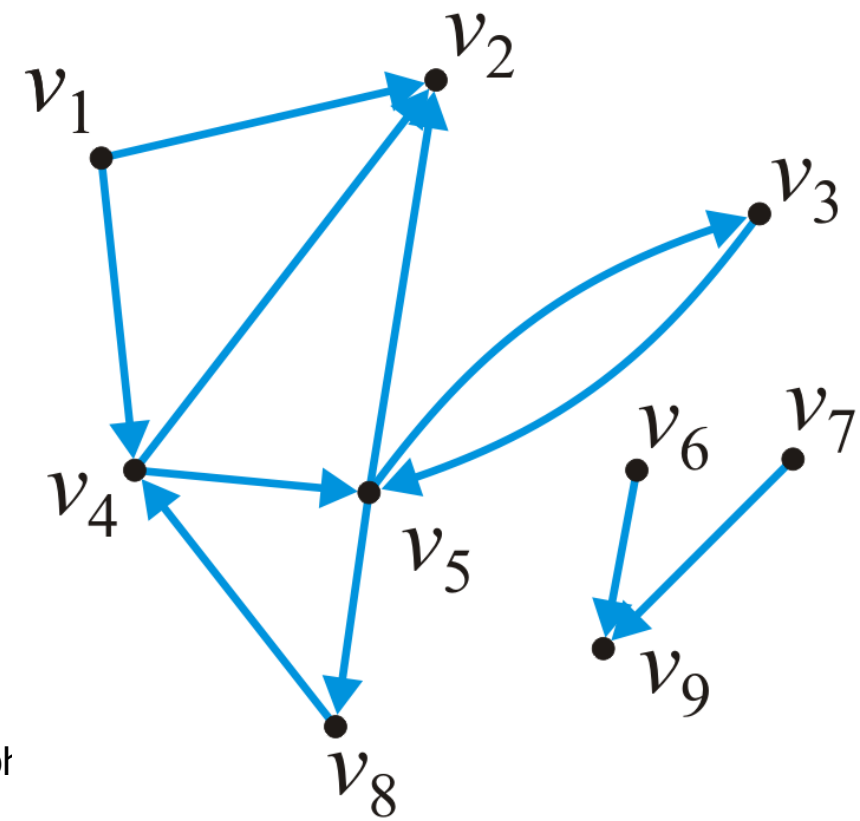
- Degree of a vertex must be modified to consider both cases:
  - Out-degree of a vertex is the number of vertices which are adjacent to the given vertex
    - Number of outgoing edges
  - In-degree of a vertex is the number of vertices which this vertex is adjacent to
    - Number of incoming edges

- In this graph:
  - In-degree(v_1) = 0 out-degree(v_1) = 2
  - In-degree(v_5) = 2 out-degree(v_5) = 3



Path

- A path in a directed graph is an ordered sequence of vertices
  - $(v_0, v_1, v_2, \ldots, v_k)$
  - where $(v_j - 1, v_j)$ is an edge for $j = 1, \ldots, k$

- A path of length 5 in this graph is
  - $(v_1, v_4, v_5, v_3, v_5, v_2)$

- A simple cycle of length 3 is
  - (v_8, v_4, v_5, v_8)